Experiment Number : 12 **Date: 03-03-26**

Sriram B

24BCS285

CSE-A1

Implementation of exception handling in Java

Aim:

Write a java program to implement exception handling.

PRE LAB EXERCISE

QUESTIONS

1. What is exception handling in Java?

- Exception handling in Java is a mechanism to handle runtime errors so that the normal flow of the program is maintained without abrupt termination.

- It uses keywords like:
  `try`, `catch`, `finally`, `throw`, `throws`.

2. What is the purpose of using try-catch blocks in exception handling?

   The purpose of try-catch blocks is to:

- Detect runtime errors
- Handle exceptions gracefully
- Prevent program crash
- Maintain normal program execution
  try → contains risky code
  catch → handles the exception if it occurs

IN LAB EXERCISE

Objective

To understand and implement exception handling through try-catch and finally blocks.

Source Code

```
import java.util.InputMismatchException;
import java.util.Scanner;
public class ExceptionHandlingExample {
 public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
         try {
        System.out.print("Enter an integer: ");
        int num = scanner.nextInt();
         int result = 10 / num;
        System.out.println("Result: " + result);
        } catch (InputMismatchException e) {
        System.out.println("Invalid input! Please enter an integer.");
        } catch (ArithmeticException e) {
        System.out.println("Cannot divide by zero!");
        } finally {
        scanner.close();
        System.out.println("Program execution completed.");
        } }
}
```

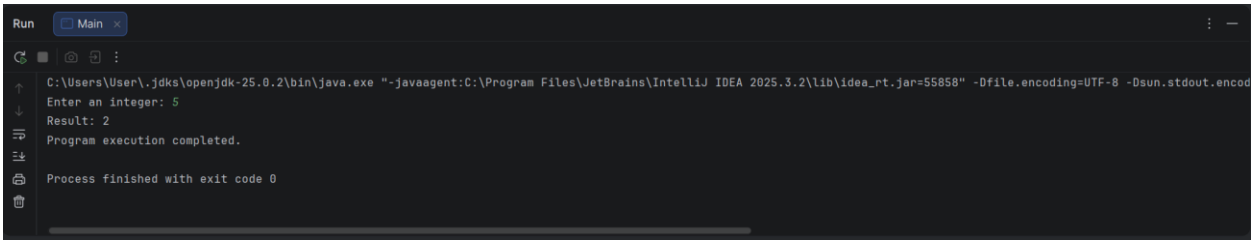
Outputs

Output 1 (Valid Input - Non-zero Integer):

Enter an integer: 5

Result: 2

Program execution completed.

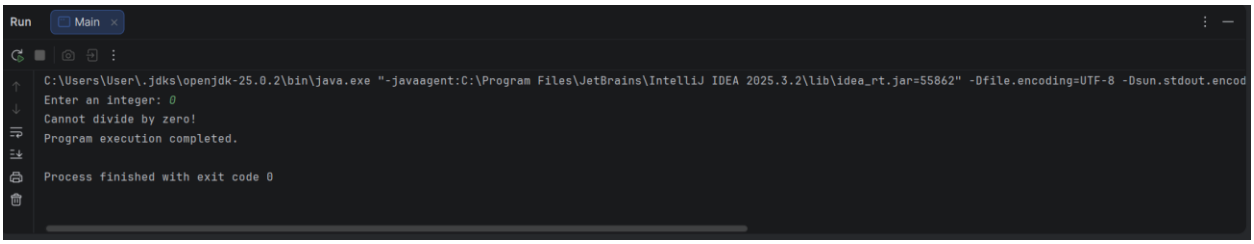


Output 2 (Valid Input - Zero):

Enter an integer: 0

Cannot divide by zero!

Program execution completed.

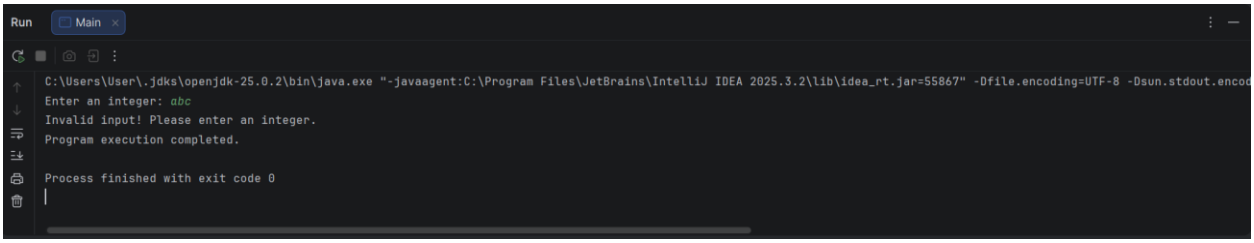


Output 3 (Invalid Input - Non-integer):

Enter an integer: abc

Invalid input! Please enter an integer.

Program execution completed.



POST LAB EXERCISE

1. What is the purpose of the Scanner object in the Java program?

   The Scanner object is used to **read input from the user** (keyboard input).
   In this program:

- Scanner scanner = new Scanner(System.in);

- It reads an integer using scanner.nextInt();
  So its purpose is to **accept user input during program execution**.

2. What exceptions are expected to be thrown in the code within the try block? Why?

Two exceptions may occur:

1. **InputMismatchException**

Occurs when:

- The user enters a value that is **not an integer** (e.g., "abc").

Reason:

- nextInt() expects an integer.
- If the input type is wrong, this exception is thrown.

---

2. **ArithmeticException**

Occurs when:

- The user enters **0**.

Reason:

- The code performs 10 / num.
- Division by zero is not allowed in Java.
- So Java throws an ArithmeticException.

So the expected exceptions are:

✓ InputMismatchException → invalid input

✓ ArithmeticException → divide by zero

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
	Faculty Signature	